

Introducing personalized decision making

THE CORE IDEA

ITEs are a rung 3 (counterfactual) concept: the difference between an individual's outcomes under two alternative interventions, such as treatment and control.

$$\text{ITE}(u) = Y_t(u) - Y_c(u)$$

vs. $\text{CATE}(c) = E[Y_t - Y_c \mid C = c]$

IS THIS VIDEO FOR ME?

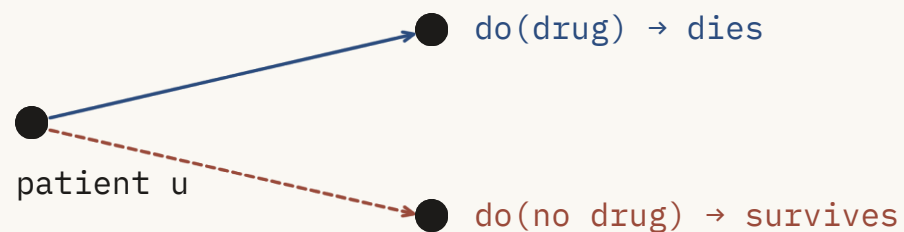
- You know causal inference and want an introduction to ITEs.
- You have read Mueller & Pearl's paper and want guidance on applying it to a practical use case.

WHAT I WILL COVER

- 01 What ITEs are, and how they differ from CATEs.
 - 02 P(benefit) and P(harm) as the counterfactual quantities of interest.
 - 03 How to process experimental and observational data to calculate those quantities.
 - 04 How to analyze them to make an informed decision on that that maximize a desired outcome.
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Equal averages can hide harm

HARM IS COUNTERFACTUAL



$$P(\text{benefit}) = P(y_t, y'_c)$$

$$P(\text{harm}) = P(y'_t, y_c)$$

A randomized trial identifies only the difference, $ATE = P(\text{benefit}) - P(\text{harm})$. Adding observational data bounds each term for a given group (Tian & Pearl, 2000).

	Women	Men
CATE, from the trial	0.28	0.28
Recovery among those who chose the drug	0.27	0.70
$P(\text{benefit})$	0.28	0.49
$P(\text{harm})$	0.00	0.21

Same CATE, different risk: about **21% of men** would survive without the drug and die with it. For women, the drug the drug harms no one.

Two notebooks, from data to decision

`ITE_Numerical_Example.ipynb`

Reproduces the paper's example

Takes the published trial and survey tables and computes the Tian–Pearl bounds on $P(\text{benefit})$ and $P(\text{harm})$ for women and men.

Result · $P(\text{harm} \mid \text{men}) = 0.21$

`ITE_Data_Generation_Example.ipynb`

Simulates the data, then estimates

Samples a trial and an observational study, checks that CATE does not differ by sex, and reports benefit and harm as intervals.

Result · $P(\text{harm} \mid \text{men}) \in [0.18, 0.23]$

ESTIMATED SURVIVAL OF 2,000 SIMULATED MEN UNDER THREE POLICIES

22.8%

Drug not approved

49.9%

Prescribed to everyone

68–73%

Withheld from those it would harm (hypothetical biomarker)

Decide for individuals

INDIVIDUAL TREATMENT EFFECT

“The difference between an individual’s outcomes under two alternative interventions, such as treatment and control.”

TWO LIMITS

1. ITEs are generally not point-identifiable: data bound them, as on slide 2. on slide 2.
2. People who share the same measured attributes get the same estimate.

WHAT IT ADDS

Each person’s own utilities and beliefs enter the decision. In high-stakes medicine, that beats optimizing optimizing a subgroup’s CATE.

28% → 62%

Effectiveness with a personalized approach, as reported by by [Mueller & Pearl \(2023\)](#)